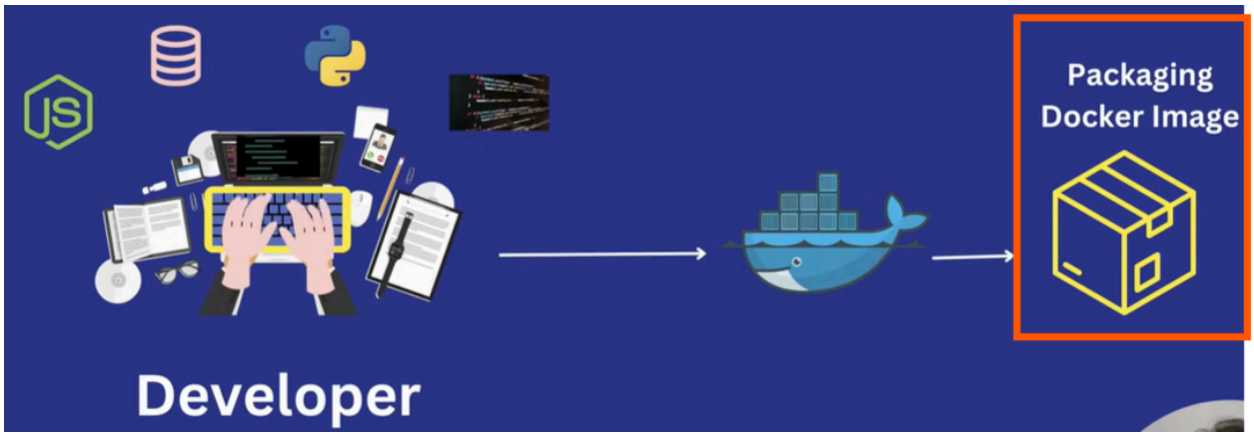
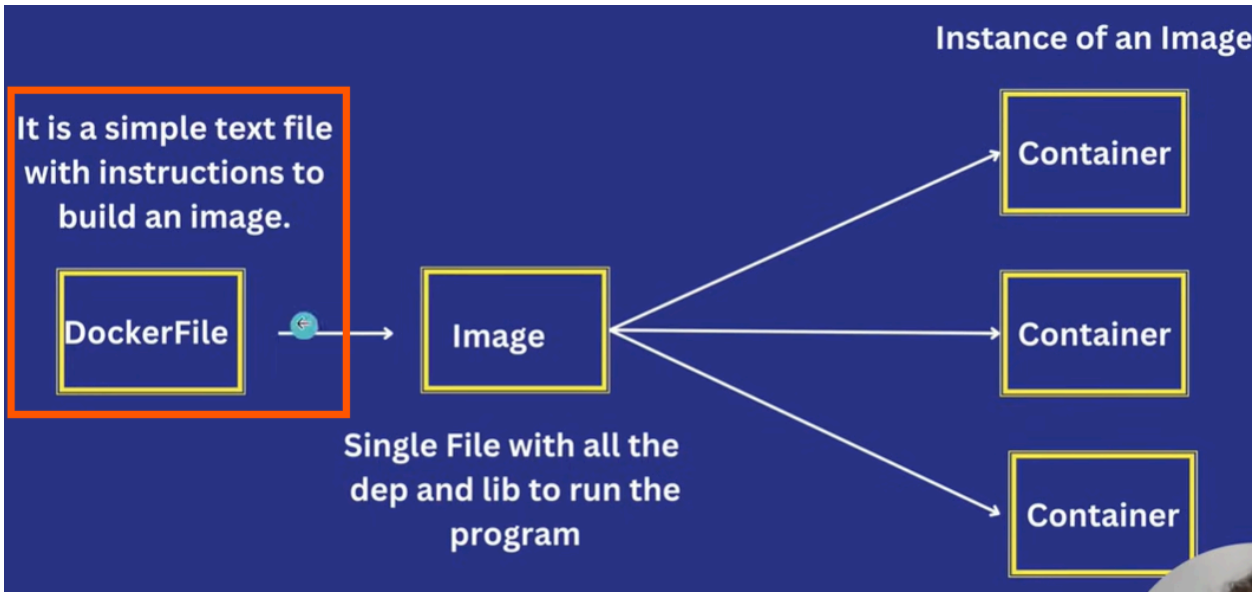
Creating Dockerfile

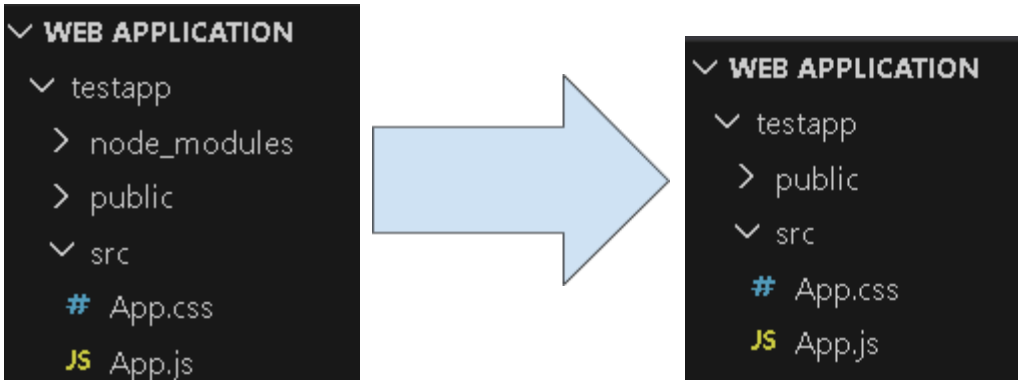
We already have demo project and docker. By using docker, need to create image of application or project.
We are going to create the image given below: in the next step



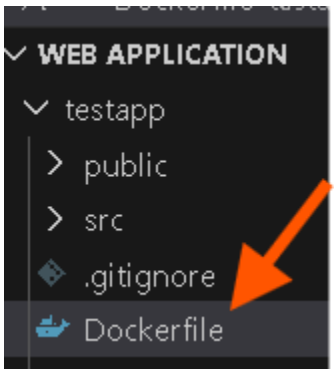
To create a image, first step is to create dockerfile,



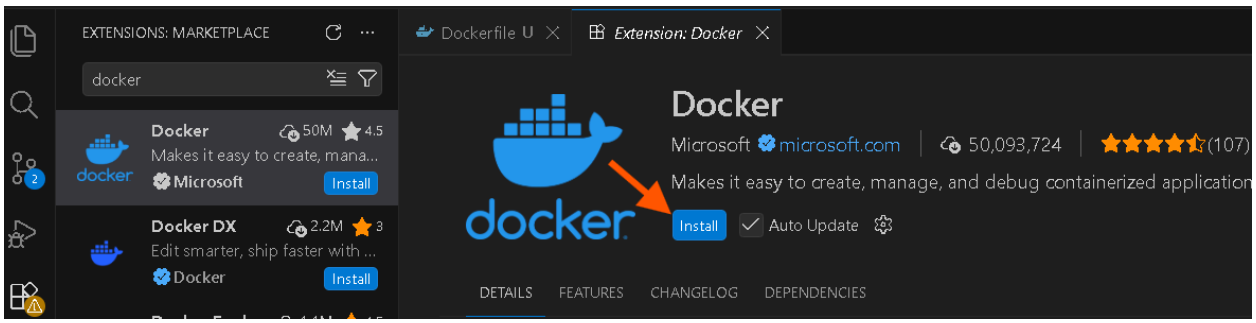
Now let's start creating a Dockerfile,
At first let's remove **node_modules** folder, due to large file size.



Create a Dockerfile, in the **testapp** folder,



To make things easier when using Docker in VsCode, install its extension.



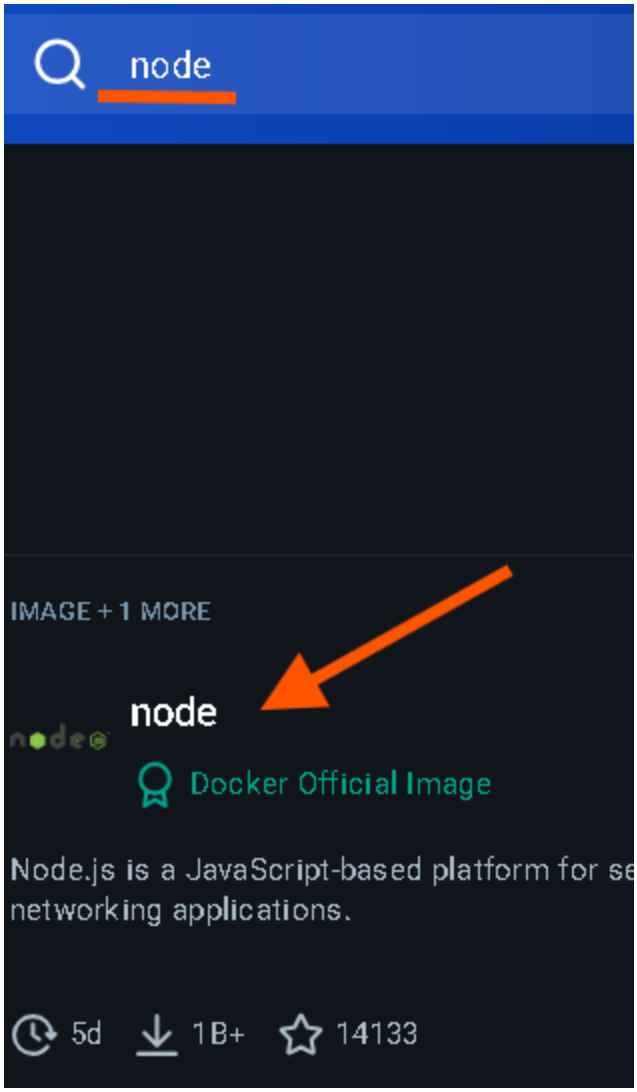
Then let's start writing Dockerfile:

We need a base to run the program, so we need node.

FROM node

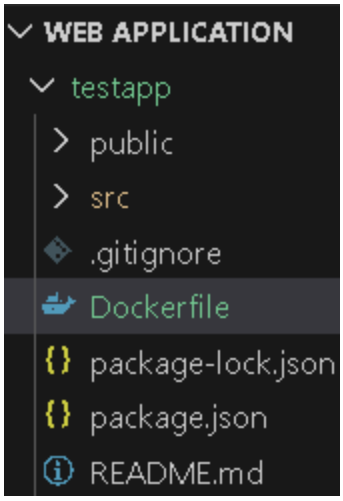
If you are writing "FROM node" meaning you are using latest version. From where this base image like node comes from. It comes from pre-defined images from Docker Hub(from where it gets download).

Search node and Select node:



if version 20 is required:

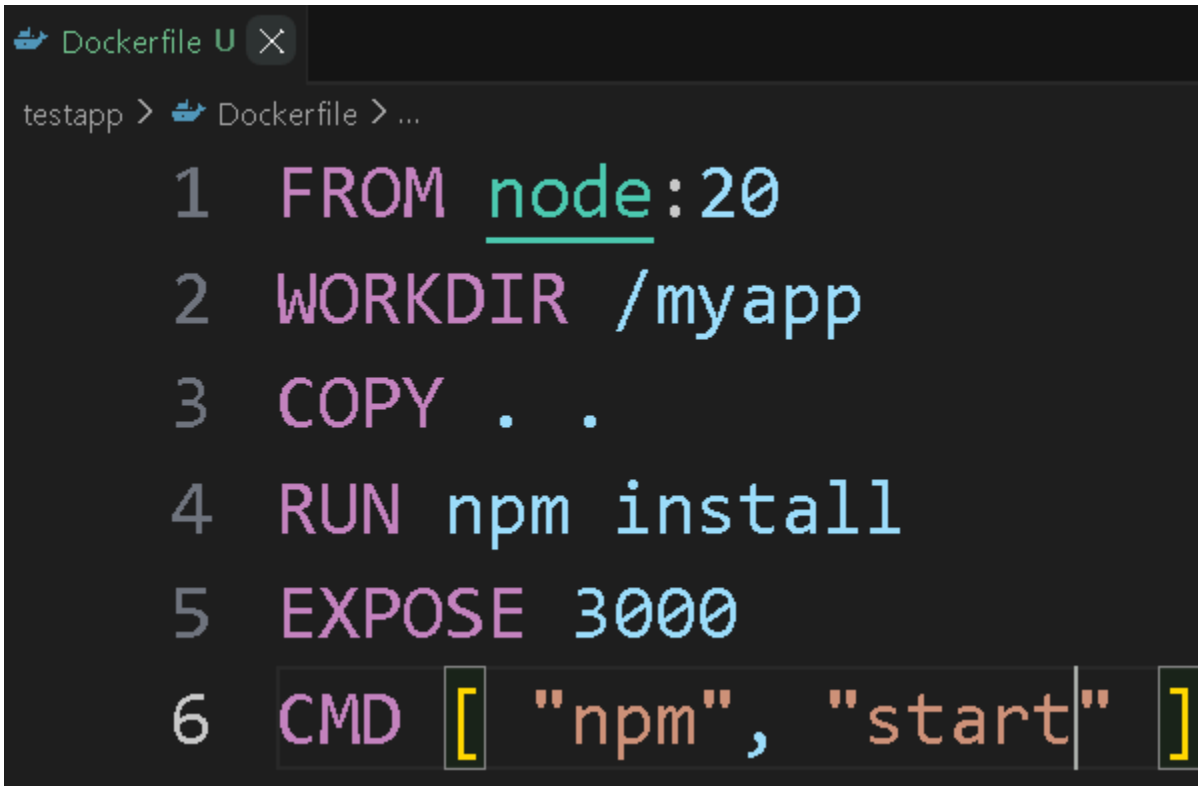
Then use the command:
FROM node:20
initially , containers are empty. Create a folder in isolated environment. Inside that folder we can run the application.
WORKDIR /myapp
WORKDIR means working directory, /myapp means creating a folder named myapp within the isolated container.
Then we have to upload the code related files. So use **COPY . .**
Or you can use **COPY . ./myapp** because we already said working directory is /myapp so you can replace with **.**
After moving the files to the container, need to run or process it.



Now to install additional folder named as node_modules. Use the command **RUN npm install.** Based on node version 20 which is mentioned in first line, will install related things in node_modules.

You can use **RUN npm start** to start the application. But main purpose is to create a image.

Now need to create a image, using **CMD ["npm", "start"]**

By default port is 3000, so web application listens to 3000. Use **EXPOSE 3000**

Where **EXPOSE 3000** is optional.